

Mohit Kori

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Professional Summary

Embedded Systems and Electronics Engineer with dual MSc degrees in Power Electronics & Control and Electronics, with hands-on experience in embedded firmware development, hardware validation, industrial electronics testing, and PCB-based system development. Skilled in Embedded C/C++, STM32, ARM Cortex-M, MATLAB/Simulink, PLC programming, debugging, and test validation. Experienced in hardware/software troubleshooting, root cause analysis, and prototype-to-validation workflows for embedded and industrial electronic systems.

Technical Skills

- **Embedded Systems & Firmware:** Embedded C/C++, STM32, ARM Cortex-M, ADC, PWM, SPI, I2C, Timers, Interrupts, UART, GPIO, Sensor Interfacing, Real-Time Embedded Systems
- **AI & Software Fundamentals:** Python, Machine Learning Fundamentals, Embedded AI Interest, TinyML Concepts, Data Analysis, Algorithmic Problem Solving
- **Automation & Control Systems:** PID Control, PLC Fundamentals, Siemens TIA Portal, Industrial Automation Concepts, Industrial Sensors, Process Control
- **Electronics & Hardware:** PCB Design, Signal Conditioning, Power Electronics, DC/DC Converters, Circuit Debugging, Hardware Validation, EMI/EMC Awareness
- **Testing & Diagnostics:** Oscilloscope, Multimeter, Functional Testing, Root Cause Analysis (RCA), Troubleshooting, Calibration, HIL Testing
- **Software & Engineering Tools:** MATLAB/Simulink, LTSpice, VS Code, KiCad, EasyEDA, Git, GitHub, MS Excel (Pivot Tables, VLOOKUP)
- **Engineering Practices:** Technical Documentation, Cross-Functional Collaboration, Structured Problem Solving, Prototype Development, System Validation, Continuous Improvement

Professional Experience

Operations Supervisor

Nov 2022 – Apr 2026

Amazon DLU2, Luton, United Kingdom

- Coordinated high-throughput logistics operations handling 5,000+ units per shift across cross-functional teams.
- Led operational troubleshooting, escalation management, and root cause analysis, improving process performance from 96% to 98%.
- Reduced operational delays by approximately 15% by analysing workflow bottlenecks, improving shift execution, and coordinating corrective actions.
- Used Excel-based KPI tracking, performance reports, and data analysis to support shift planning, resource allocation, and continuous improvement.

Electronics & Electrical Test Engineer

Jan 2021 – Sep 2022

Multispan Control Instruments Pvt. Ltd., Ahmedabad, India

- Tested and validated 50+ industrial electronic systems including embedded control boards, power supply units, and DC/DC converter modules.
- Performed PCB-level fault diagnosis, circuit debugging, and electrical troubleshooting using oscilloscopes, multimeters, and hardware analysis tools.
- Improved fault detection efficiency by approximately 20% through structured debugging, test validation, and root cause analysis workflows.
- Supported hardware validation, prototype verification, and engineering testing across embedded and industrial electronics products.
- Produced 30+ technical reports, test documentation, and quality records supporting compliance, traceability, and product validation.

Engineering Projects

STM32-Based Line-Following & Obstacle-Aware System

- Developed embedded firmware in Embedded C/C++ on STM32 ARM Cortex-M platform for real-time motor and sensor control.
- Implemented ADC-based sensor acquisition, interrupt-driven logic, and PWM motor control reducing response delay by 18%.
- Applied PID control algorithm improving tracking stability and navigation performance.
- Performed hardware/software debugging using oscilloscopes and structured root cause analysis.

Autonomous Hazard Inspection Robot

- Developed autonomous embedded navigation system using sensors and real-time obstacle detection logic.
- Built MATLAB/Simulink models for system validation and hardware-in-loop testing.
- Conducted simulation-to-hardware verification and iterative firmware optimisation.

Custom PCB Design — KiCad & EasyEDA

- Designed schematic and PCB layouts for microcontroller-based embedded systems.
- Applied signal routing, DRC validation, and power integrity principles to improve hardware reliability.

Education

MSc in Power Electronics and Control University of Hertfordshire, United Kingdom	2023
MSc in Electronics Sardar Patel University, India	2020
BSc in Instrumentation Sardar Patel University, India	2018

Certifications & Additional Information

Certification: Siemens TIA Portal PLC Programming
Languages: English (Fluent), German (A2), Hindi (Native)
Availability: Immediate Joining Available
Work Authorization: Germany Opportunity Card